Guideline

PERI-OPERATIVE MANAGEMENT OF PATIENTS WITH ADVANCED CHRONIC KIDNEY DISEASE (CKD) UNDERGOING PARATHYROIDECTOMY

Reference # MMG028

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Renal medicine, dialysis and endocrinology	Medical, nursing and pharmacy

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Introduction[1]

Patients with advanced CKD may be considered for a subtotal parathyroidectomy. Post-operative hypocalcaemia is the most common complication following parathyroidectomy. Any deviation on the guidelines set below will require a Consultant-to-Consultant discussion.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

Definitions

CKD	Chronic Kidney Disease
Ca	Calcium

PO_4	Phosphate
Mg	Magnesium
iPTH	Intact parathyroid hormone
CVC	Central Venous Catheter
IV	Intravenous
CNS	Clinical Nurse Supervisor
TDS	Three times each day
BD	Twice daily
UEC	Urea, electrolytes, creatinine
HD	Haemodialysis
PD	Peritoneal dialysis

Clinical presentation of Hypocalcaemia[1]

Patients with mild hypocalcaemia are often asymptomatic. Patients with acute, moderate to severe hypocalcaemia may present with neuromuscular irritability which manifest as peri-oral paraesthesia, muscle twitching and spasm. Moderate to severe hypocalcaemia can also cause cardiovascular instability leading to hypotension, cardiac arrhythmia, or heart failure. Acute severe hypocalcaemia can be a life-threatening emergency and can cause tetany, laryngospasm, and seizures.

Management Strategies[1-3]

Preoperative medication adjustments by the Surgical team:

- Continue regular phosphate binders and cease cinacalcet (if applicable)
- Commence or increase dose of **calcitriol** to **2mcg (8 x 0.25mcg capsules) daily in divided doses** for one week prior to surgery unless the adjusted serum calcium is >2.7mmol/L on a recent blood test (within 7 days).
- Once a surgical date is set, please inform the patient, renal team, and dialysis CNS to arrange for preoperative dialysis planning (home or hospital).

Day of Surgery - Parathyroidectomy

- Check biochemistry on day of surgery for: albumin-corrected Ca, PO_4, iPTH, Mg, 25-hydroxy-vitamin D, UECs
- Intraoperative insertion of CVC should be considered in selected patients who are:
    - at high risk for developing postoperative hypocalcaemia,
    - have poor venous access, and;
    - have anticipated difficulties with frequent postoperative blood sampling and likely to require urgent IV medication administration of calcium.

Postoperative medications to be charted by the Medical Officer:

- Oral Calcitriol: initiate at 0.5mcg (2 x 0.25mcg capsules) TDS then increase, if required, to a maximum of 1.5mcg (6 x 0.25mcg capsules) TDS.
- Oral Calcium: 1000mg to 1200mg TDS spaced away from meals
    - increase, if required or as tolerated, to a maximum of 3000mg to 3600mg TDS
- Phosphate binders: calcium or non-calcium, prescribed **with meals** if serum PO_4 >1.6mmol/L.
- Oral Magnesium: magnesium aspartate 500mg (1.64mmol) BD if hypomagnesaemic.

Post-operative biochemical monitoring schedule

	Investigations required:	**Monitoring required:**
Immediately postop:	Ca, Mg, iPTH, EUC	

Postoperative:	Ca, PO_4, Mg, EUC	Check bloods 6 hourly for 48 hours post-operatively.
48 hours postoperatively until discharge OR 12hrs post cessation IV Calcium	Ca, Mg, PO_4, EUC	Check bloods twice daily until stable

***EUC (electrolytes, urea, creatinine)**

Indications for IV Calcium gluconate infusion:

- Symptomatic hypocalcaemia
- **Corrected Serum Ca**:
  - <1.8mmol/L **and/or**
  - 1.8 – 2.1 mmol/L and symptomatic **and/or**
  - >2.1mmol/L and falling rapidly (>10% in the preceding 4-6 hours)

See Appendix 1 for full infusion protocol

FOLLOW UP POST DISCHARGE

Requirements for calcium supplements and calcitriol upon discharge are likely to vary considerably between individual patients and ***must be closely monitored***.

Post discharge biochemical monitoring schedule

Post discharge– 1 week	Ca, PO_4, EUC	Predialysis bloods 3x/week until Ca is within the normal range
1 week – 1 month	Ca, PO_4	At least weekly until within normal range
1 month – 2 months	Ca, PO_4	At least fortnightly unless outside the normal range
2 months -12 months	Ca	At least monthly unless outside the normal range

For Home Haemodialysis patients: The **first HD** session post discharge must be organised in the incentre unit. Results and patients must be medically reviewed prior to discharge from the incentre HD Unit.

Peritoneal dialysis and renal transplant patients: Arrange biochemical monitoring with Home Therapies, Senior Renal Registrar or usual Renal Physician.

Practical Points

Availability: calcium gluconate (2.2mmol calcium) 10mL vial

Related Documents

Legislation:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016

Authorising Policy and Standard/s:

- MP 077/18 Statewide Medicines Formulary Policy
- SCGOPHCG Medicines Management Policy and Procedure

References

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2. NFK-KDOQI Guidelines. Guideline 15. Parathyroidectomy in Patients with CKD
3. Cozzolino M, Gallieni M, Corsi C, Bastagli A, BrancaccioD. Management of calcium refilling post-parathyroidectomy in end-stage renal disease. J Nephrol 2004; 17: 3-8
4. eTG complete. Endocrinology 2016 Available from : https://www-tg-org-au.qelibresources.health.wa.gov.au/

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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Appendix 1: Intravenous calcium gluconate infusion protocol[4]

Indications:

- Symptomatic hypocalcaemia
- **Corrected Serum Ca** is:
  - <1.8mmol/L **and/or**
  - 1.8 – 2.1 mmol/L and symptomatic **and/or**
  - >2.1mmol/L and falling rapidly (>10% in the preceding 4-6 hours)

- **Calcium gluconate** is the preferred agent over calcium chloride due to reduced toxic effects on peripheral veins. This **must be administered** using an infusion pump via a **CVC** *unless* otherwise discussed with the patient's usual nephrologist (or consultant on call).

INITIAL (LOAD) IV CALCIUM TREATMENT (Must be followed by maintenance infusion as per below)		
Calcium gluconate monohydrate (931mg/10mL) (equivalent to an elemental Ca load of 2.2 mmol/10mL)		
Indication	***Prescription***	***Follow up***
Only if corrected serum Ca <1.8 mmol/L and/or symptomatic hypocalcemia	20mL (4.4mmol) of calcium gluconate in 50mL Normal Saline over 30 minutes[4]	Repeat Ca in 2 hours post bolus or earlier if necessary

IV CALCIUM MAINTENANCE INFUSION		
60mL (13.2mmol) calcium gluconate (2.2mmol Ca/10mL) in 250mL of Normal Saline (total volume 310mL). **Note: Volume does not need to be removed from this bag prior to reconstitution**		
Initial infusion rate	20 mL/hr (=0.85mmol/hr)	Repeat Ca in 2 hours

FOLLOW UP INFUSION RATES BASED ON SERUM CALCIUM MONITORING

Corrected serum Ca:	***Prescription***		***Follow up***
<1.8mmol/l and/or symptomatic hypocalcaemia	Re-load with 20mL (4.4mmol) of calcium gluconate in **50mL** Normal Saline over 30 minutes	**AND** Increase maintenance infusion rate to 30 mL/hr **AND** increase oral calcium supplements	Repeat Ca in 2 hours
1.8 - 2.1mmol/l	30ml/hr **AND** increase oral calcium and/or calcitriol m supplements		Repeat Ca in 4 hours
2.1 - 2.2mmol/l	Continue at current rate **AND** continue oral calcium and/or calcitriol supplements		Repeat Ca in 4 hours
> 2.2mmol/l	Cease infusion **AND** continue oral calcium and/or calcitriol supplements		Repeat Ca in 4 hours